

Jay Patel

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SUMMARY

AI/ML Engineer with 5+ years of experience in designing, developing, and deploying scalable machine learning and deep learning solutions. Proficient in Python, TensorFlow, PyTorch, and scikit-learn, with expertise in NLP, Predictive Modeling, and Intelligent Automation. Skilled in building end-to-end ML pipelines, automating model workflows with Airflow and MLflow, and deploying production-grade solutions using Docker and NVIDIA Triton Inference Server. Strong background in data preprocessing, feature engineering, and model optimization, coupled with experience in data visualization and reporting using Power BI. Adept at collaborating with cross-functional teams to translate complex data into actionable insights and drive measurable business improvements.

SKILLS

Programming Languages: Python, R, SQL
Machine Learning & Deep Learning Frameworks: TensorFlow, Keras, PyTorch, scikit-learn, XGBoost, LightGBM, Hugging Face Transformers
Python Packages: pandas, NumPy, SciPy, Matplotlib, Seaborn, spaCy, NLTK, Sentence-Transformers, OpenCV
Algorithms & Techniques: Random Forest, XGBoost, Gradient Boosting, Logistic Regression, Decision Trees, LSTM, CNN, BERT, RoBERTa, LayoutLM, Sentence-BERT, K-Means, PCA, ARIMA, NER, Text Classification, OCR (Tesseract), Feature Engineering, Hyperparameter Tuning
MLOps, Automation & Deployment: MLflow, Apache Airflow, NVIDIA Triton Inference Server, Docker, FastAPI, AWS ECS, Prometheus, Continuous Retraining, Model Monitoring
Cloud & Data Storage: AWS S3, AWS ECS, SQL Server
Data Engineering & Pipelines: ETL Development, Data Preprocessing, Data Integration, Data Cleaning, Model Deployment Pipelines
Visualization & Reporting: Power BI, Matplotlib, Seaborn, ggplot2 (R)
Soft Skills & Collaboration: Cross-Functional Collaboration, Model Interpretation, KPI Definition, Performance Optimization, Documentation, Mentoring, Agile Practices

EXPERIENCE

State Street, USA AI/ML Engineer	Oct 2024 - Present
- Designed and implemented an AI-driven Intelligent Document Processing (IDP) platform to automate extraction of structured data from financial reports, invoices, and fund statements using NLP, OCR, and deep learning, improving document processing accuracy by 42%. - Built and fine-tuned Transformer-based models (BERT, LayoutLM, RoBERTa) using PyTorch and Hugging Face Transformers for Named Entity Recognition (NER), text classification, and key-value pair extraction. - Developed scalable data pipelines for preprocessing and annotation using Python, pandas, NumPy, and spaCy, standardizing input from over 1.5M unstructured documents. - Implemented OCR extraction with Tesseract and applied semantic text embeddings using Sentence-BERT to enhance entity linking and contextual understanding. - Deployed optimized models on AWS ECS using Docker and NVIDIA Triton Inference Server, achieving 35% lower inference latency and 3× higher throughput for large-scale document batches. - Automated model retraining and monitoring pipelines with MLflow (model tracking), Airflow (workflow orchestration), and Prometheus (performance monitoring), enabling continuous model lifecycle management. - Integrated outputs into SQL Server and Power BI dashboards, providing real-time insights on document extraction quality, confidence scores, and accuracy metrics for business analysts and auditors.	
Hexaware Technologies, India Machine Learning Engineer	Mar 2019 - Jun 2023
- Designed and developed a Predictive Maintenance ML solution to forecast equipment failures using Python, TensorFlow, and scikit-learn, improving maintenance scheduling accuracy by 35% and reducing downtime costs. - Collected, cleaned, and processed over 20M sensor data points from IoT devices using PySpark and AWS S3, ensuring data quality and pipeline reliability. - Built and optimized feature engineering workflows leveraging pandas, NumPy, and SQL, improving model accuracy by 12% through advanced feature selection and scaling techniques. - Developed supervised learning models (Random Forest, XGBoost) for anomaly detection and remaining useful life (RUL) prediction, tuning hyperparameters with GridSearchCV for best performance. - Deployed trained models as REST APIs using FastAPI and Docker, integrated into client's existing ERP system for real-time inference and alert generation. - Implemented MLOps practices with MLflow for model versioning and Airflow for pipeline automation, reducing retraining time from 2 hours to 20 minutes. - Created interactive Power BI dashboards to visualize equipment health metrics and prediction trends, enhancing transparency for maintenance and operations teams.	

EDUCATION

Master of Science in Computer Science Pace University New York, USA	May 2025
Bachelor of Computer Applications Veer Narmad South Gujarat University Surat, India	Feb 2019

CERTIFICATIONS

• Career Essentials in Data Analysis by Microsoft and LinkedIn • BCG - GenAI Job Simulation